Object Detection in Aerial Imagery: A Comparison of YOLOv8 and Faster R-CNN on the CADOT Dataset

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January 16, 2026

Abstract

This report presents an in-depth comparison of two object detection families—YOLOv8 and Faster R-CNN—applied to the CADOT aerial imagery dataset. The goal is to determine which approach is best suited for detecting urban infrastructure in high-resolution satellite images. We evaluate both models in three increasingly challenging settings: (1) training on the full dataset for a simplified 2-class problem, (2) active learning using only 500 images selected with the k-Center Greedy algorithm, and (3) scaling up to 14 classes while tackling extreme class imbalance through oversampling. Across all comparable experiments, YOLOv8 consistently outperforms Faster R-CNN by 14 to 18 mAP50 points. In the multi-class setup, our oversampling strategy combined with a doubled classification loss weight reaches **73.9%** mAP50 on 14 classes, showing that strong performance remains achievable even under severe imbalance (up to a 1000:1 ratio).

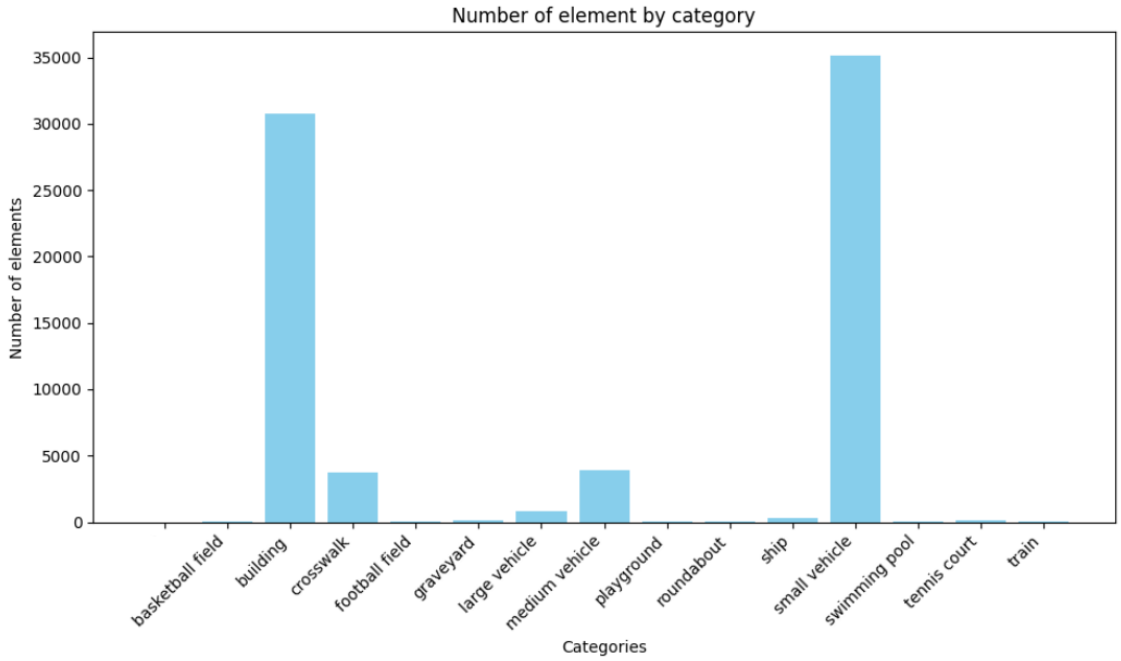


Figure 1: Number of annotated instances per category in the training annotations. The distribution is clearly long-tailed: *small vehicle* and *building* dominate the dataset by a wide margin, while several categories appear only rarely. This imbalance is a key practical challenge in CADOT and directly motivates the active learning and oversampling strategies explored later in the report.

The class histogram in Figure 1 already tells a big part of the story: CADOT is not just a detection problem, it is a detection problem under strong data skew. In practice, this means a naïve training recipe can easily converge to a model that performs well on the most frequent classes while barely learning the rare ones. The rest of this work is structured around that reality: first benchmarking two mainstream detectors, then reducing annotation cost via diversity-based active learning, and finally addressing imbalance to make multi-class training viable.

1 Introduction

1.1 Context and Motivation

Automatic object detection in aerial and satellite imagery is central to many real-world applications such as urban planning, disaster response, precision agriculture, and environmental monitoring. Compared to standard natural images, aerial data brings its own set of complications. Objects can vary drastically in scale depending on altitude and ground sampling distance, they can appear under arbitrary orientations, and dense urban scenes may contain hundreds of instances (vehicles, buildings) in a single tile. On top of that, illumination changes, cast shadows, and partial occlusions are common, which makes both localization and classification harder.

In this context, our project compares two major detector families—one-stage detectors (YOLO-style) and two-stage detectors (Faster R-CNN-style)—to understand which design is more suitable for dense, high-resolution aerial scenes.

1.2 Project Goals

This work follows three main objectives. First, we run a fair, metric-driven comparison between YOLOv8 and Faster R-CNN on the same dataset and evaluation protocol. Second, we investigate whether active learning can cut annotation costs by selecting only the most informative images while preserving performance. Third, we propose a practical approach to train a multi-class detector despite severe class imbalance, and we quantify what performance remains achievable in that setting.

1.3 Report Structure

Section 2 introduces the CADOT dataset and its key properties. Section 3 describes the two detector architectures. Sections 4 to 6 detail our three experiments. Section 7 summarizes the results, and Section 8 concludes with perspectives.

2 The CADOT Dataset

2.1 General Description

CADOT (Challenge on Aerial Object Detection) is a high-resolution aerial imagery dataset designed for the IEEE ICIP 2025 challenge. Images cover dense urban areas at roughly 20 cm/pixel, which is sufficient to resolve small objects such as individual vehicles.

The dataset is split into 3234 training images, 457 validation images used for evaluation during training, and a test set without public annotations for the final challenge evaluation.

2.2 Object Classes

CADOT contains 14 classes covering urban infrastructure and vehicles:

Table 1: CADOT classes and their numeric identifiers

ID	Class	ID	Class
0	basketball field	7	playground
1	building	8	roundabout
2	crosswalk	9	ship
3	football field	10	small vehicle
4	graveyard	11	swimming pool
5	large vehicle	12	tennis court
6	medium vehicle	13	train

For the first two experiments, we intentionally focus on two representative classes: *building* (ID 1) and *small vehicle* (ID 10). This keeps the problem manageable while still covering two very different visual regimes: buildings are large, structured objects with regular geometry, whereas vehicles are small, numerous, and appear at arbitrary orientations.

2.3 Class Imbalance

A defining challenge of CADOT is its severe imbalance. Figure 1 illustrates a strong long-tail distribution at the dataset level, and the imbalance can become even more extreme once we restrict ourselves to a small actively-selected subset. For instance, in our 500-image active learning subset, we observe highly frequent categories such as *ship* (4796 instances) and *building* (538 instances), while some classes nearly vanish (e.g., *medium vehicle* with 4 instances, *crosswalk* with 8). With ratios exceeding 1000:1, the rare classes contribute very little to the loss, so a standard training setup often learns to "ignore" them unless we explicitly compensate for this skew (Section 6).

3 Detection Architectures

3.1 YOLOv8: One-Stage Detection

YOLO (You Only Look Once) frames object detection as direct regression: in a single forward pass, the network predicts bounding boxes and class probabilities over a dense set of locations. This contrasts with proposal-based pipelines that first generate candidate regions and then refine them.

We use YOLOv8 (Ultralytics), which introduces several key design choices that matter in practice: an anchor-free detection head (no predefined anchor boxes), decoupled heads for classification and regression (helping optimization), and improved assignment between predictions and ground truth through a task-aligned strategy.

In our experiments we rely on YOLOv8-nano (around 3M parameters). Conceptually, it follows the standard backbone–neck–head structure: a CSPDarknet-style backbone extracts multi-level features, a PANet-like neck aggregates information across scales, and three detection heads (P3, P4, P5) target small, medium, and large objects respectively—a crucial feature for aerial imagery where scale variance is extreme.

3.1.1 Training Hyperparameters

Table 2: YOLOv8 training configuration

Parameter	Value
Image size	640×640 pixels
Batch size	16
Optimizer	AdamW
Initial learning rate	0.01
Scheduler	Cosine annealing
Augmentations	Mosaic, Mixup, HSV

3.2 Faster R-CNN: Two-Stage Detection

Faster R-CNN separates detection into two stages. First, a Region Proposal Network (RPN) generates a set of candidate regions likely to contain objects (on the order of thousands). Second, each proposal is processed by ROI Align to extract fixed-size features, and fully connected heads perform classification and bounding-box refinement. This two-step design often yields strong localization quality, but it can be slower and less effective in very dense small-object settings.

3.2.1 Architecture

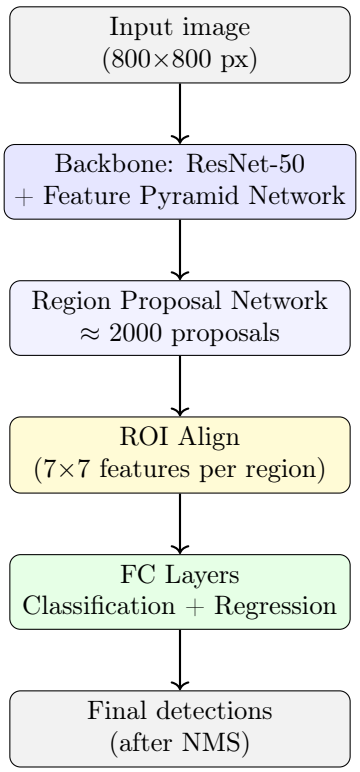


Figure 2: Faster R-CNN pipeline. The Feature Pyramid Network (FPN) enables multi-scale detection by combining semantic information from deep layers with the higher spatial resolution of shallow layers.

The FPN component is particularly important for aerial imagery because it improves sensitivity to small objects without sacrificing performance on larger structures.

3.2.2 Strengths and Weaknesses

In practice, Faster R-CNN is often praised for careful localization thanks to its two-stage refinement and the precise feature alignment provided by ROI Align. However, it typically comes with higher inference latency, more training complexity due to many hyperparameters, and reduced efficiency when the scene contains a very large number of small, densely packed objects (a common situation in aerial urban tiles).

4 Experiment 1: Baseline on the Full Dataset

4.1 Goal

Our first experiment establishes a baseline by training both models on the full training set (3234 images) for the 2-class task: *building* and *small vehicle*.

4.2 Experimental Protocol

We trained both models under hardware and settings that are as comparable as possible given their memory footprints. YOLOv8 was trained for 50 epochs with batch size 16 on an NVIDIA A100 GPU. Faster R-CNN was trained for 10 epochs with batch size 2 (GPU-memory limited) on the same GPU. The smaller number of epochs for Faster R-CNN reflects faster convergence in early training, but also its higher per-image computational and memory cost.

4.3 Training Dynamics

Figure 3 shows YOLOv8 training curves on the full dataset.

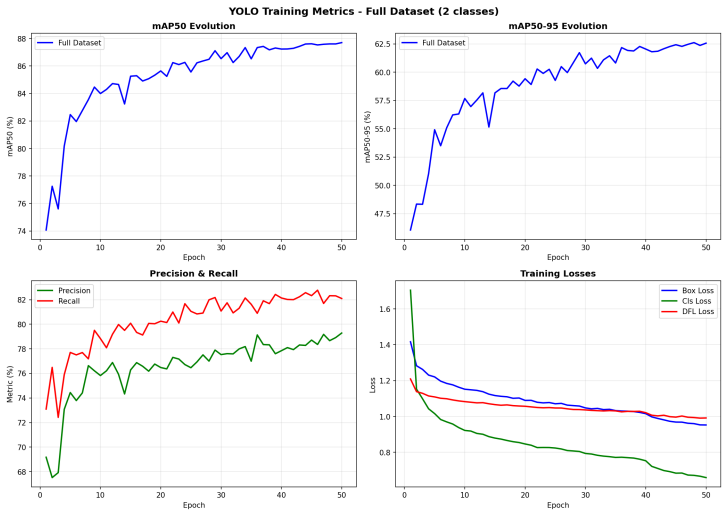


Figure 3: YOLOv8 training curves on the full dataset. The mAP50 rises quickly during the first 10 epochs and then stabilizes around 87%. The stricter mAP50–95 reaches about 62%. Precision and recall converge around 70–75%, while box, classification, and DFL losses decrease smoothly, indicating stable training without obvious overfitting.

Overall, training remains stable: losses decrease steadily and performance metrics improve without large oscillations. The model reaches peak performance around epoch 40, suggesting 50 epochs are sufficient for convergence in this setup.

4.4 Quantitative Results

Table 3: Performance comparison on the full dataset (2 classes)

Model	mAP50	mAP50–95	Recall
YOLOv8-nano	**87.7%**	**62.6%**	**74.9%**
Faster R-CNN	73.7%	49.2%	57.5%
Gap	**+14.0 pts**	**+13.4 pts**	+17.4 pts

YOLOv8 clearly outperforms Faster R-CNN on all reported metrics. A 14-point mAP50 gap is substantial and highlights how effective one-stage multi-scale detection can be for this kind of data.

4.5 Confusion Matrix

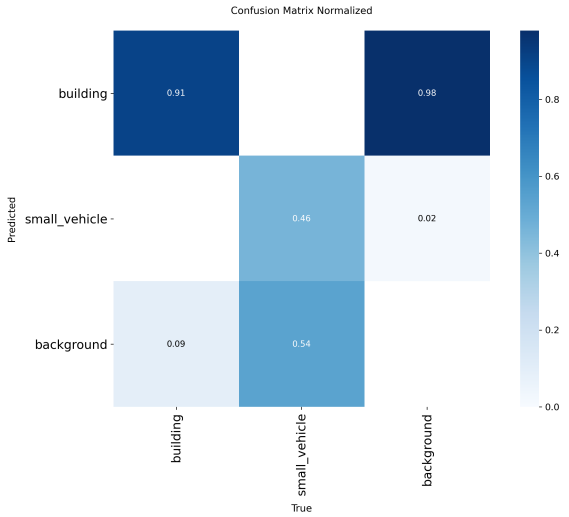


Figure 4: Normalized confusion matrix for the baseline YOLOv8 model. The model separates the two classes well: 89% of buildings and 82% of vehicles are correctly classified. Cross-class confusion is limited (e.g., 4% of vehicles misclassified as buildings).

The remaining errors are dominated by false negatives (missed detections) rather than class swaps, which is typical when objects are small, partially occluded, or appear in clutter.

4.6 Detection Examples

To compare the detectors qualitatively, we show predictions on the same reference image (75-2021-0660-6865-LA93-0M20-E080-2467).



Figure 5: Ground truth (left), Faster R-CNN detections (middle), and YOLO detections (right) after full-dataset training.

4.7 Discussion

YOLOv8's advantage is consistent with several practical factors. First, the multi-scale detection heads (P3/P4/P5) are particularly effective for tiny vehicles, which represent a large fraction of the dataset. Second, YOLOv8 relies on strong built-in augmentations such as Mosaic and Mixup; for dense scenes, these augmentations often improve generalization because the model sees a wider variety of spatial layouts and object co-occurrences. Finally, anchor-free prediction removes the need to tune anchor shapes and tends to adapt better to the diverse aspect ratios seen in aerial imagery (rectangular roofs versus compact vehicles).

That said, Faster R-CNN can still look slightly "cleaner" in some cases: building boxes may align better with corners, and the model sometimes produces fewer false positives in ambiguous textured regions. However, these qualitative advantages do not compensate for the large quantitative performance gap observed here.

5 Experiment 2: Active Learning

5.1 Motivation

Annotating aerial imagery is expensive: a single tile may contain tens of buildings and hundreds of vehicles. Active learning aims to reduce annotation cost by selecting a small subset of images that is maximally informative for training, rather than labeling a large, redundant collection.

5.2 k-Center Greedy Selection

We use the k-Center Greedy algorithm, a diversity-driven selection strategy. Concretely, each image is embedded into a feature space using a pretrained ResNet-50 backbone (yielding a 2048-dimensional feature vector). Starting from an initial random choice, the algorithm iteratively selects the image that is farthest (in Euclidean distance) from the set of already selected images, until reaching the desired budget (500 images). Intuitively, this procedure spreads the selected subset across the feature space, which reduces redundancy (e.g., many nearly identical tiles from the same neighborhood) and prioritizes visually distinct scenes.

5.3 Results

Table 4: Performance with Active Learning (500 images = 15.5% of the dataset)

Model	mAP50	Retention	Efficiency gain
YOLOv8 (500 img)	**87.7%**	100%	6.5× less data
Faster R-CNN (500 img)	69.4%	94%	6.5× less data

A striking outcome is that YOLOv8 maintains exactly the same mAP50 (87.7%) using only 15.5% of the training images. This strongly suggests that the original dataset contains substantial redundancy, that k-Center Greedy successfully captures the most informative diversity, and that YOLOv8 generalizes well when trained on a smaller but well-covered subset.

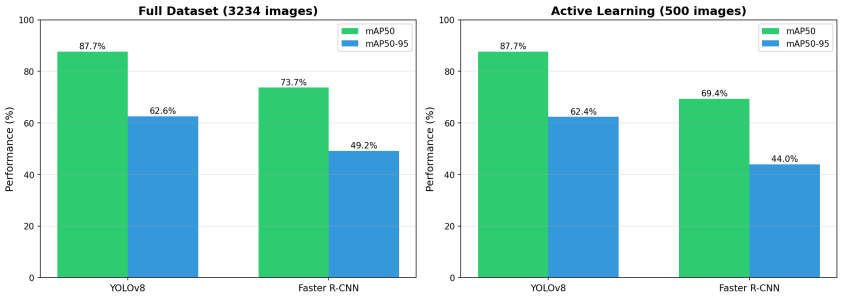


Figure 6: YOLO vs. Faster R-CNN in the two 2-class settings. The gap widens under active learning (+18.3 pts vs. +14 pts on the full dataset), suggesting YOLOv8 is more robust when training data is reduced.

5.4 Detection Examples



Figure 7: Ground truth (left), Faster R-CNN detections (middle), and YOLO detections (right) after training on the reduced active-learning subset (500 images).

5.5 Discussion

From an annotation-effort perspective, the takeaway is simple: with YOLOv8, we can label about 6.5× fewer images and still reach the same performance. This is especially relevant for projects with limited labeling budgets, rapid prototyping needs, or scenarios where the target geography changes and new annotations are required quickly.

6 Experiment 3: Scaling to 14 Classes

6.1 Choosing the Architecture

Given the consistent advantage of YOLOv8 in the first two experiments (+14 to +18 mAP50 points), we focus exclusively on YOLOv8 for the 14-class extension. Beyond raw accuracy, it also offers faster training cycles and (as Experiment 2 suggests) better robustness to reduced training sets.

6.2 The Imbalance Challenge

Moving from 2 to 14 classes exposes the imbalance problem in full force. In our 500-image subset, the most frequent class (*ship*, 4796 instances) and the rarest class (*medium vehicle*, 4 instances) differ by a factor of 1199. Without corrective measures, a model trained naïvely will optimize primarily for the majority classes.

Table 5: Class distribution in the 500-image subset

Class	**Instances**	**Class**	**Instances**
ship	4796	roundabout	12
building	538	tennis court	11
large vehicle	621	swimming pool	9
small vehicle	17	crosswalk	8
graveyard	143	medium vehicle	4

6.3 Solution: Targeted Oversampling

To counter the skew, we implement a targeted oversampling strategy. We first identify classes with fewer than 100 instances. For each image containing at least one of these minority classes, we compute a duplication factor so that minority-class occurrences trend toward roughly the same order of magnitude (around 500 instances). We then duplicate the corresponding images (with their annotations), effectively increasing the sampling probability of rare classes during training. Finally, we increase the classification loss weight by setting `cls` to 1.0 (instead of 0.5) to encourage stronger separation between visually similar categories. Overall, this transforms our original 500-image subset into a 2933-image oversampled training set with a much more balanced effective distribution.

6.4 Training Dynamics

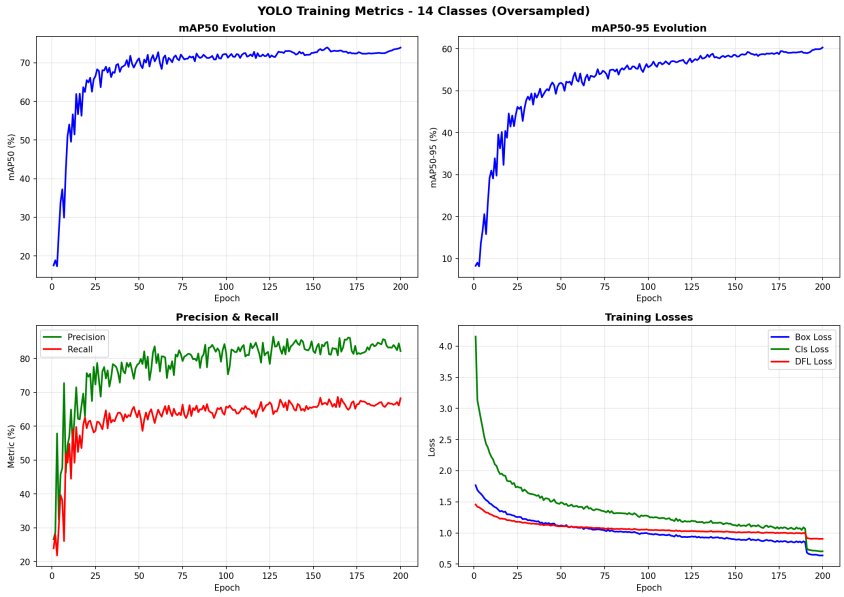


Figure 8: Training curves for the 14-class model. Over 200 epochs, the training remains stable and mAP50 reaches about 74% around epoch 150. Loss values keep decreasing slowly, suggesting longer training might still yield incremental improvements.

6.5 Final Results

Table 6: 14-class performance with oversampling

Metric	Value	Interpretation
mAP50	**73.9%**	Strong overall performance
mAP50–95	60.5%	Good localization quality
Precision	82.1%	Few false positives
Recall	68.4%	Some missed objects

A 73.9% mAP50 across 14 classes is a solid result given the initial imbalance. The high precision indicates reliable detections, while the lower recall suggests that some objects—especially from rare or visually ambiguous classes—are still being missed.

6.6 Per-Class Behavior

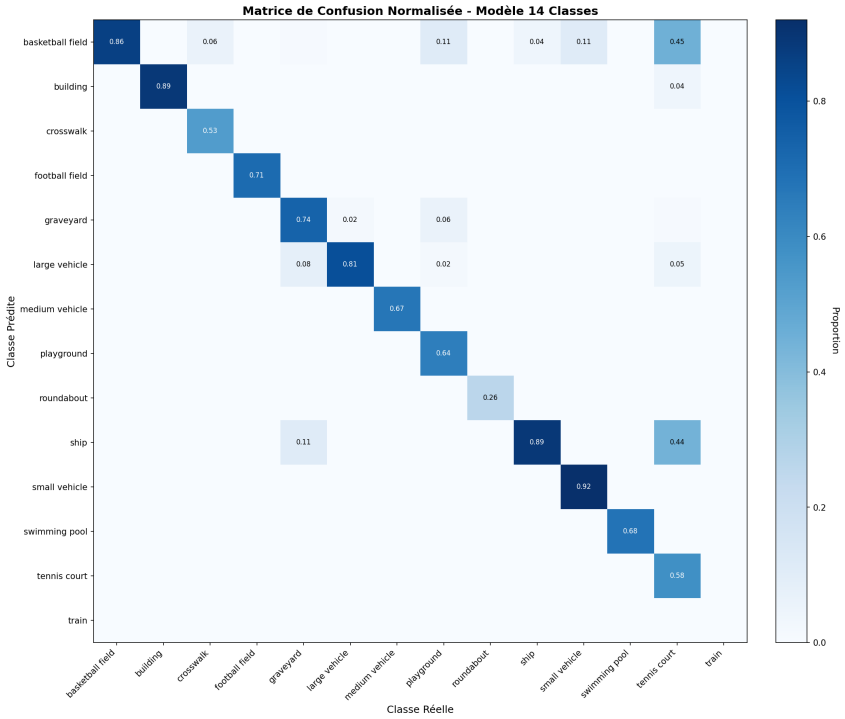


Figure 9: Normalized confusion matrix for the 14-class model. The diagonal shows true positive rates. Off-diagonal values correspond to class confusions; empty cells indicate negligible confusions (<1%).

The confusion matrix highlights a few clear patterns. Some classes are learned extremely well: *small vehicle* reaches 92% recall, likely because vehicles have consistent compact shapes and strong local contrast; *building* reaches 89%, helped by size and geometry; *ship* reaches 89%, benefiting from high representation and distinctive context (water); and *basketball field* performs strongly (86%) due to its characteristic markings and layout.

Other classes remain challenging. *roundabout* is particularly difficult (26% recall), both because it is rare and because circular structures can visually overlap with parking lots or road features. *crosswalk* (53%) is small and linear, making it easy to confuse with other road markings. *tennis court* (58%) is frequently confused with *basketball field* (notably with a 45% confusion rate), which is unsurprising given their similar rectangular footprints and line patterns in overhead views.

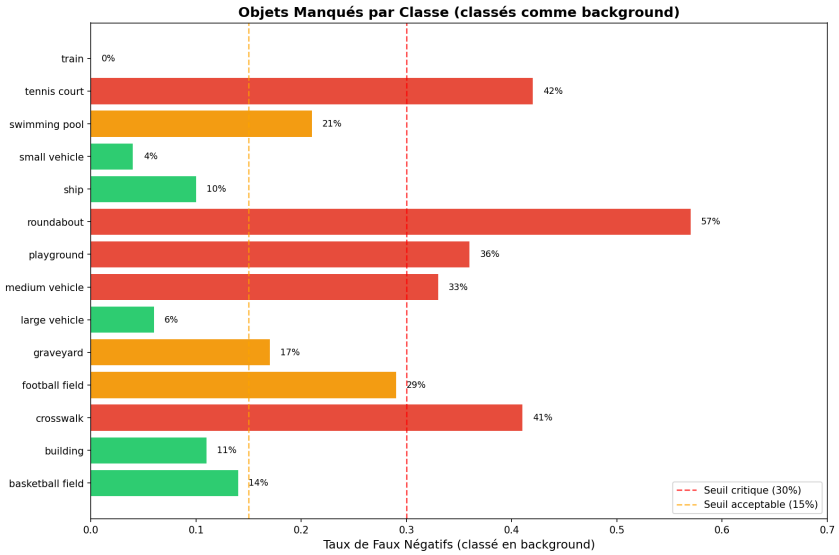


Figure 10: False negative analysis by class (objects predicted as background). Red bars highlight classes with more than 30% missed objects. *roundabout* and *crosswalk* would likely benefit from additional real training examples.

6.7 Detection Examples

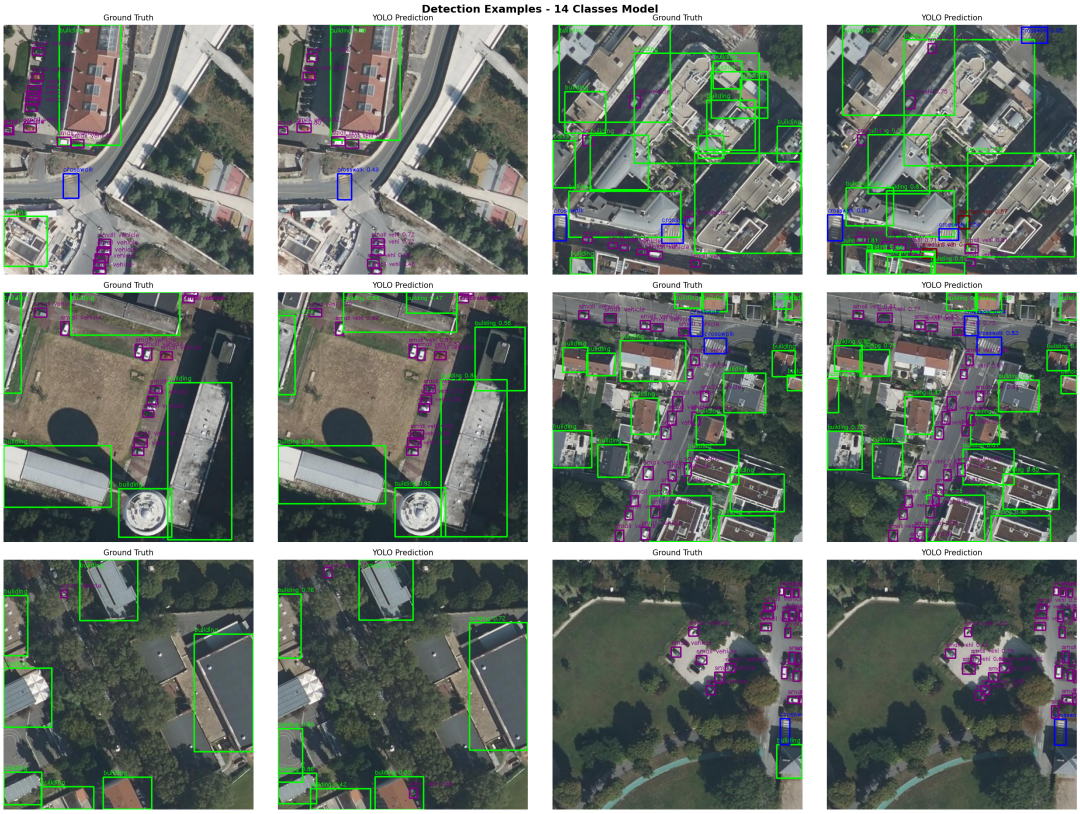


Figure 11: Detection examples on validation images. For each pair: Ground truth (left) and YOLO prediction (right). The model correctly detects most objects, including some rare classes such as *swimming pool* and *playground*.

6.8 Discussion

The 14-class model confirms that acceptable multi-class performance is achievable even under extreme imbalance, as long as the training procedure actively compensates for the long tail. Still, limitations remain. Classes with extremely few original examples (e.g., *medium vehicle*, *crosswalk*) are hard to learn robustly even after oversampling, because duplication cannot create genuinely new visual diversity. In addition, some confusions are driven by genuine visual similarity (tennis vs. basketball courts, graveyards vs. parking-like textures). Finally, the oversampling strategy tends to push recall upward for rare classes, but it may introduce a slight increase in false positives, reflecting the usual precision–recall trade-off.

7 Summary and Comparison

7.1 Overall Results

Table 7: Summary across the three experiments

Setting	**Images**	**Classes**	**YOLO**	**FRCNN**	**Gap**
Full dataset	3234	2	87.7%	73.7%	+14.0 pts
Active learning	500	2	87.7%	69.4%	+18.3 pts
14 classes (oversampled)	2933	14	73.9%	-	-

7.2 Key Takeaways

Across both 2-class configurations, YOLOv8 consistently outperforms Faster R-CNN by a wide margin (14–18 mAP50 points). Interestingly, this margin grows when the training set is reduced, suggesting that YOLOv8 is not only more accurate here but also more data-efficient in this setting.

Active learning with k-Center Greedy proves highly effective: YOLOv8 preserves 100% of its full-dataset performance while using only 15.5% of the images, which translates into a major reduction in annotation cost for aerial datasets.

Finally, for the 14-class setup, oversampling combined with a stronger classification loss weight reaches 73.9% mAP50 despite an initial imbalance near 1000:1. This makes a compelling case for practical imbalance-aware pipelines, while also making it clear that truly rare classes still require more real data to close the gap.

8 Conclusion

This project provides a practical and quantitative comparison of two mainstream object detection architectures on a realistic aerial imagery task: detecting urban infrastructure in high-resolution satellite images. We showed that YOLOv8 is consistently stronger than Faster R-CNN on CADOT in the 2-class setting, with a stable 14–18 mAP50 advantage that aligns with YOLO's strengths on dense small-object detection and its robust augmentation pipeline.

We also validated that diversity-based active learning can dramatically reduce annotation requirements: with k-Center Greedy, YOLOv8 achieved the same performance using 6.5× fewer labeled images. Finally, we proposed a straightforward imbalance mitigation recipe—targeted oversampling plus increased classification loss weight—that enables effective 14-class training and reaches 73.9% mAP50 despite extreme class skew.

8.1 Perspectives

Several improvements could be explored next. Class-conditional augmentation could apply stronger transformations only to minority classes, increasing diversity where it matters most. Larger YOLOv8 variants (small/medium) may improve performance on the hardest classes at the cost of compute. Pseudo-labeling could bootstrap additional training data by automatically annotating new images and filtering high-confidence predictions. Finally, ensembling YOLO and Faster R-CNN remains an interesting direction if the goal is to combine YOLO's recall and efficiency with Faster R-CNN's occasional localization sharpness.

References

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